Weighted Lucas Sums and Differences: Coefficient Classification and Four OEIS Sequences

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Abstract

We classify positive representations of $L_i - qL_j$, $qL_i - L_j$, and $qL_i + L_j$ for integer q. For $q \geq 2$, the first form has only finitely many collisions, with an explicit uniform index bound. The other two have infinitely many repeated values precisely for three arithmetic coefficient families: integral Lucas quotients, integral Fibonacci quotients, and half-Lucas coefficients. Signed quotients of Lucas gaps characterize all finite collisions, and global uniqueness can be decided by a finite search. For $q = 5$, we give direct all-index formulas for OEIS A191980–A191983, classify all repeated values and pairwise intersections, and determine exact counting functions and growth rates.

2020 Mathematics Subject Classification. Primary 11B39; Secondary 11B50, 05A15.

Keywords. Lucas numbers; Fibonacci numbers; representation collisions; linear recurrences; counting functions.

1 Introduction

Representations of integers by sums and differences of terms of linear recurrences arise naturally in enumeration and Diophantine questions. The Fibonacci and Lucas sequences provide an especially tractable setting because their closed forms, algebraic conjugates, and addition identities control differences between indices; see Koshy [1]. Here we study positive values of weighted Lucas sums and differences, in particular when different index pairs yield the same value.

We first establish uniform finite collision bounds and classify coefficients admitting infinitely many collisions. For the coefficient $q = 5$ we then obtain direct terms, multiplicities, intersections, and counting asymptotics for four OEIS sequences [2, 3, 4, 5].

Notation. Let $F_0 = 0, F_1 = 1$ and $L_0 = 2, L_1 = 1$, with $u_{n+2} = u_{n+1} + u_n$. Put $\varphi = (1 + \sqrt{5})/2$, $\alpha = \varphi$, and $\beta = -\alpha^{-1}$, so that Binet's formula is $L_n = \alpha^n + \beta^n$. For negative indices use $F_{-n} = (-1)^{n+1} F_n$ and $L_{-n} = (-1)^n L_n$. Put

$$S_q = \{L_i - qL_j > 0 : i, j \geq 1\}, \quad T_q = \{qL_i - L_j > 0 : i, j \geq 1\}, \quad P_q = \{qL_i + L_j > 0 : i, j \geq 1\}.$$

In a representation $L_i - qL_j$, $qL_i - L_j$, or $qL_i + L_j$, we call i the *first index* and j the *second index*. A collision is an equality of values from distinct ordered index pairs. The recurrence gives

$$L_{m+4} - L_m = 5F_{m+2}, \qquad F_a L_b = F_{a+b} + (-1)^b F_{a-b}. \tag{1}$$

2 Infinite-collision constructions

Theorem 2.1 (Lucas–Fibonacci quotient coefficients)**.** *Let $m > h \geq 1$ have the same parity, and suppose*

$$q = \begin{cases} L_m/L_h \in \mathbb{Z}, & m, h \text{ odd}, \\ F_m/F_h \in \mathbb{Z}, & m, h \text{ even}. \end{cases}$$

Then $q \geq 2$, and for every $n \geq 1$,

$$qL_{n+m-h} - L_n = qL_{n+m+h} - L_{n+2m} > 0. \tag{2}$$

In particular, T_q and P_q each contain infinitely many repeated positive values.

Proof. For m odd, the Lucas product formula gives

$$L_{n+2m} - L_n = L_m L_{n+m}, \qquad L_{n+m+h} - L_{n+m-h} = L_h L_{n+m}.$$

For m, h even, the Lucas–Fibonacci identity instead gives

$$L_{n+2m} - L_n = 5F_m F_{n+m}, \quad L_{n+m+h} - L_{n+m-h} = 5F_h F_{n+m}.$$

In either case dividing the two identities gives

$$L_{n+2m} - L_n = q(L_{n+m+h} - L_{n+m-h}),$$

which is (2). The expression $qL_{n+m-h} - L_n$ on the left-hand side of (2) is positive and strictly increasing with n, since $q \geq 2$ and $m - h \geq 1$. The equivalent sum collision is

$$qL_{n+m-h} + L_{n+2m} = qL_{n+m+h} + L_n.$$

□

Theorem 2.2 (Half-Lucas collision coefficients)**.** *For every integer $r \geq 0$, let $M = 6r + 3$ and*

$$q = \frac{L_M}{2} \in \mathbb{Z}.$$

Then T_q and P_q each have infinitely many repeated positive values. More precisely, for every $x \geq M$,

$$qL_x - L_{x-M+1} = qL_{x+3} - L_{x+M+1} > 0, \tag{3}$$

and

$$qL_x + L_{x+M+1} = qL_{x+3} + L_{x-M+1}.$$

The coefficients begin $2, 38, 682, 12238, 219602, \ldots$.

Proof. The Lucas residues modulo 2 have period 3, so L_M is even when $3 \mid M$. As M is odd, the product identity gives

$$L_{x+M+1} - L_{x-M+1} = L_M L_{x+1}.$$

The recurrence also gives $L_{x+3} - L_x = 2L_{x+1}$. Substituting $2q = L_M$ proves (3) and the sum identity. As $x \geq M \geq 3$, $qL_x - L_{x-M+1} > 0$; these values strictly increase with x. □

3 Norm separation and finite collision bounds

Lemma 3.1 (Algebraic-norm separation)**.** *For integers $d \geq 1$ and $q \geq 1$, with $\alpha = \varphi$ and $\beta = -\alpha^{-1}$,*

$$|\alpha^d - q| \geq \frac{1}{q+1}. \tag{4}$$

Proof. The conjugate algebraic integers $\alpha^d - q$ and $\beta^d - q$ have nonzero integral product: $\alpha^d = F_d\alpha + F_{d-1}$ is irrational. Since $|\beta^d - q| \leq q + 1$, the bound follows. □

For completeness, this estimate also makes every bounded value search finite. If $0 < L_i - qL_j \leq X$, then $i > j$ and Binet's formula gives

$$L_j \leq (q+1)(X+q+1)+1, \qquad L_i \leq X + qL_j.$$

The bounds for $0 < qL_i - L_j \leq X$ follow by exchanging the indices when $j > i$, and by $(q-1)L_i \leq X$ when $j \leq i$ and $q \geq 2$. For sums, $qL_i + L_j \leq X$ directly bounds both indices. Thus for any integer coefficient, all distinct values at most X, their multiplicities (when finite), and their ranks are determined by a finite search. The case of negative coefficients reduces to the positive case by exchanging sum and difference forms.

Proposition 3.2 (No simultaneous-shift positive collisions)**.** *For every integer $q \geq 2$, there are no indices $i < k$, $j < \ell$ such that*

$$L_{n+i} - qL_{n+j} = L_{n+k} - qL_{n+\ell} > 0 \quad \textit{for every } n \geq 0.$$

Proof. An equality for all shifts implies, by Binet's formula, both its α- and β-coefficient equalities. Put $h = k - i \geq 1$, $e = \ell - j \geq 1$, and $s = i - j$. Positivity for large n forces $\alpha^s > q$, so $s \geq 2$, and

$$\alpha^s(\alpha^h - 1) = q(\alpha^e - 1)$$

forces $e > h$, hence $e \geq 2$. Conjugation gives

$$q|\beta^e - 1| = |\beta|^s|\beta^h - 1| \leq \alpha^{-2}(1 + \alpha^{-1}) = \alpha^{-1}.$$

But $q|\beta^e - 1| \geq 2(1 - \alpha^{-2}) = 2/\alpha > \alpha^{-1}$, a contradiction. □

4 The global finite-collision bound

The preceding bounds determine all representations below a given value X. The next result gives a uniform index bound for collisions in S_q, independent of X.

Theorem 4.1 (Uniform complete collision certificate)**.** *Fix an integer $q \geq 2$. Let $t = t(q)$ be the least positive index such that*

$$L_t > 200(q+1)^4. \tag{5}$$

If two distinct positive representations satisfy

$$L_i - qL_j = L_k - qL_\ell > 0, \qquad i < k,$$

then

$$\boxed{k \leq 6t.} \tag{6}$$

Consequently every S_q with $q \geq 2$ has only finitely many multiply represented positive values. Its complete collision list and global uniqueness can be determined by checking the finitely many pairs $1 \leq j < i \leq 6t$.

Proof. Write $\alpha = \varphi$, $\beta = -\alpha^{-1}$. The equality forces $j < \ell$. Moreover, positivity gives $k > \ell$. Put $d = k - \ell$, $h = k - i$ and $g = k - j$. We prove $k \le 6t$ by contradiction. The bounds

$$\frac{1}{2}\alpha^n \le L_n \le 2\alpha^n \quad (n \ge 0), \qquad L_k - L_i \ge L_{k-2}$$

follow from Binet's formula and monotonicity. Since

$$L_{k-2} \le L_k - L_i = q(L_\ell - L_j) < qL_\ell,$$

we have $\alpha^d < 4q\alpha^2 < \alpha^t$; thus $1 \le d < t$. Suppose $k > 6t$. Then $\ell = k - d > 5t$.

Write

$$x = L_k - qL_\ell = \alpha^\ell(\alpha^d - q) + \beta^k - q\beta^\ell.$$

The norm argument in (4) gives $|\alpha^d - q| \ge 1/(q+1)$. Since $|\beta^k - q\beta^\ell| \le q + 1$ and $\alpha^\ell > 2(q+1)^2$, the positivity of x forces $\alpha^d - q > 0$ and

$$x \ge \frac{\alpha^\ell}{2(q+1)}.$$

But $x < L_i \le 2\alpha^i$, so $\alpha^{\ell-i} < 4(q+1) < \alpha^t$, which gives $1 \le h < 2t$.

Next, the original equality can be rearranged as

$$qL_j = L_i - L_k + qL_\ell = \alpha^k A + \beta^k A',$$

where

$$A = \alpha^{-h} - 1 + q\alpha^{-d}, \qquad A' = \beta^{-h} - 1 + q\beta^{-d}.$$

Both are conjugate algebraic integers, and

$$|A'| \le \alpha^h + 1 + q\alpha^d < 3\alpha^{2t} < \alpha^{3t}.$$

If $A = 0$, then $qL_j = |\beta^k A'| < 1$, an impossibility. Otherwise AA' is a nonzero integer, so $|A| \ge \alpha^{-3t}$. As $k > 6t$ and $qL_j > 0$, this forces $A > 0$ and

$$qL_j > \tfrac{1}{2}\alpha^{k-3t}.$$

Using $qL_j \le 2q\alpha^j$ gives $\alpha^{k-j} < 4q\alpha^{3t} < \alpha^{4t}$, hence $g < 4t$.

Finally, express the collision as

$$0 = L_k - L_i - qL_\ell + qL_j = \alpha^k E + \beta^k E',$$

with the conjugate algebraic integers

$$E = 1 - \alpha^{-h} - q\alpha^{-d} + q\alpha^{-g}, \quad E' = 1 - \beta^{-h} - q\beta^{-d} + q\beta^{-g}.$$

The bounds on d, h, g imply

$$|E'| \le 1 + \alpha^{2t} + q\alpha^t + q\alpha^{4t} < \alpha^{6t}.$$

If $E \ne 0$, the integral norm $EE' \ne 0$ gives $|E| \ge 1/|E'|$. The displayed equality now implies $\alpha^{2k} \le |E'|^2 < \alpha^{12t}$, contrary to $k > 6t$.

If $E = 0$, the equality also forces $E' = 0$. It then holds after every simultaneous translation of all four indices. Because $\alpha^d - q > 0$, the translated differences are positive for all sufficiently large translations, contradicting Proposition 3.2. Thus neither case is possible, and $k \le 6t$. □

5 The stationary collision bound

Theorem 5.1 (Finite stationary pattern criterion)**.** *Fix any integer $q \geq 2$ and choose t as in (5). If*

$$qL_k - L_\ell = qL_i - L_j > 0, \qquad i < k, \tag{7}$$

and $k > 12t$, then all four indices lie in $[k - 6t, k + t]$, and the same equality holds after every *simultaneous nonnegative shift of the indices. Consequently, the set T_q has infinitely many multiply represented positive values if and only if there are indices with*

$$K = 12t + 1, \quad K - 3t \leq i < K, \quad K - t \leq \ell \leq K + t, \quad K - 6t \leq j < \ell$$

such that

$$0 < qL_K - L_\ell = qL_i - L_j, \qquad qL_{K+1} - L_{\ell+1} = qL_{i+1} - L_{j+1}. \tag{8}$$

This is a finite, exact, necessary-and-sufficient integer test for the existence of infinitely many positive collisions in T_q. Every collision not belonging to a stationary family has larger first index $k \leq 12t$.

Proof. Equation (7) forces $j < \ell$ and $q(L_k - L_i) = L_\ell - L_j$. The bounds $L_{n-2} \leq L_n - L_{n-1}$ and $\frac{1}{2}\alpha^n \leq L_n \leq 2\alpha^n$ yield

$$qL_{k-2} \leq L_\ell, \qquad L_{\ell-2} \leq qL_k.$$

Consequently, with $\delta = \ell - k$,

$$\alpha^{-\delta} < 4\alpha^2/q < \alpha^t, \qquad \alpha^\delta < 4q\alpha^2 < \alpha^t,$$

so $|\delta| < t$. Assume $k > 12t$. Set

$$x = qL_k - L_\ell = \alpha^k A + \beta^k A', \quad A = q - \alpha^\delta, \quad A' = q - \beta^\delta.$$

The conjugate algebraic integers A, A' have nonzero integral norm, including when $\delta = 0$ (since then $A = q - 1$). As $|A'| < \alpha^{2t}$, we have $|A| > \alpha^{-2t}$. Positivity of x and $k > 12t$ imply $A > 0$ and $x > \frac{1}{2}\alpha^{k-2t}$. Since $x < qL_i \leq 2q\alpha^i$, the gap $h = k - i$ satisfies $0 < h < 3t$.

Next,

$$L_j = qL_i - qL_k + L_\ell = \alpha^k B + \beta^k B',$$

where $B = q\alpha^{-h} - q + \alpha^\delta$ and $B' = q\beta^{-h} - q + \beta^\delta$. The conjugates are algebraic integers and $|B'| < \alpha^{5t}$. If $B = 0$, then $L_j < 1$, a contradiction; otherwise their integral norm gives $|B| > \alpha^{-5t}$. As $k > 12t$ and $L_j > 0$, we obtain $L_j > \frac{1}{2}\alpha^{k-5t}$. Thus $g = k - j < 6t$. Since $j < \ell$, also $g > -t$.

Now the collision is

$$0 = qL_k - qL_i - L_\ell + L_j = \alpha^k E + \beta^k E',$$

where

$$E = q - q\alpha^{-h} - \alpha^\delta + \alpha^{-g}, \quad E' = q - q\beta^{-h} - \beta^\delta + \beta^{-g}.$$

Both are algebraic integers and $|E'| < \alpha^{8t}$. If $E \neq 0$, the integral norm gives $|E| > 1/|E'|$ and the equality forces $\alpha^k \leq |E'| < \alpha^{8t}$, contradicting $k > 12t$. Therefore $E = E' = 0$, and the identity holds under all simultaneous shifts. The positive leading coefficient A makes its common values positive and unbounded under increasing shifts.

Every infinite sequence of distinct collision values must have unbounded k, and so contains a stationary identity of the preceding form. Translate its indices to $k = K = 12t + 1$. The established gap bounds give precisely the finite ranges in (8). The leading coefficient $A > 0$ ensures positivity at K. Conversely, the two equalities in (8) give consecutive zeros of a Lucas-recursive difference. They therefore give the identity at all shifts, and its positive values become arbitrarily large. This proves the criterion and the asserted bound for nonstationary collisions. □

Proposition 5.2 (Finite certificates for all three forms)**.** *Fix $q \geq 2$ and t as in (5). If T_q has no stationary collision, every positive collision of T_q or P_q has all indices at most $13t$. If a stationary collision exists, both forms have a positive collision with all indices at most $13t+1$. In either case global uniqueness is decided by a finite search. For S_q, the bound is $6t$. For $q = 1$ all three forms have infinitely many collisions; for $q = 0$, T_0 is empty and S_0, P_0 have infinite representation multiplicities. For $q = -m < 0$, T_q is empty, $S_q = P_m$, and $P_q = S_m$ after exchanging index roles.*

Proof. Theorem 4.1 settles S_q. When T_q has no stationary collision, Theorem 5.1 puts its larger first index at most $12t$. Positivity gives $L_j < qL_i$, and hence $j < i + t$, so all indices are at most $13t$. A sum collision with unequal first indices has a corresponding signed difference

$$v = qL_k - L_j = qL_i - L_\ell.$$

If $v > 0$, this is a collision in T_q; if $v < 0$, its negative is one in S_q. The case $v = 0$ would give two different representations of the same Lucas ratio $q > 1$, contrary to Lemma 7.1 below. Thus the same index bound holds for P_q. If T_q has a stationary collision, Theorem 5.1 supplies one at $k = 12t + 1$ with all other indices at most $13t + 1$; swapping subtracted indices gives the required sum collision. The remaining signs follow from the recurrence and the definitions. □

6 Explicit classification of infinite collisions

The quotient and half-Lucas identities exhaust the infinite-collision coefficients. We classify the index gaps of every stationary identity.

Lemma 6.1 (Rigidity of the two index gaps)**.** *Suppose $q \geq 2$, $i < k$, $j < \ell$, and*

$$q(L_{n+k} - L_{n+i}) = L_{n+\ell} - L_{n+j} \qquad (n \geq 0). \tag{9}$$

Set $h = k - i$, $e = \ell - j$, and $d = i - j$. The only possibilities are:

1. *$h = 2r$, $e = 2m$, $m > r \geq 1$, with $m \equiv r \pmod 2$, $d = m - r$, and either $q = L_m/L_r$ for m, r odd or $q = F_m/F_r$ for m, r even;*
2. *$h = 1$, $e = 2m$, $m \geq 3$ odd, $d = m + 1$, and $q = L_m$;*
3. *$h = 3$, $e = 2m$, $m \geq 3$ odd and divisible by 3, $d = m - 1$, and $q = L_m/2$;*
4. *$(h, e, d, q) = (1, 3, 2, 2)$ or $(2, 3, 0, 2)$.*

Every identity on this list is realized, with positive values and arbitrarily large simultaneous shifts.

Proof. Let $\alpha = \varphi$, $\beta = -\alpha^{-1}$, and set

$$C(s) = L_s - 1 - (-1)^s, \qquad \Phi(s) = \frac{1 + \alpha^{-s}}{1 - \alpha^{-s}} \quad (s \geq 1).$$

The two conjugate Binet coefficient equations of (9) give

$$q\alpha^i(\alpha^h - 1) = \alpha^j(\alpha^e - 1), \qquad q\beta^i(\beta^h - 1) = \beta^j(\beta^e - 1).$$

Their norms yield

$$q^2 C(h) = C(e), \qquad i \equiv j \pmod 2. \tag{10}$$

In particular $e > h$, and $d = i - j$ is even. Dividing the two Binet equations, and using $\beta = -\alpha^{-1}$, gives the exact relation

$$\alpha^{2d} = \alpha^{e-h} \frac{\Phi(h)^{[h \text{ odd}]}}{\Phi(e)^{[e \text{ odd}]}}. \tag{11}$$

The function $\Phi(s)$ decreases strictly to 1; direct calculation gives

$$\Phi(1) = \alpha^3, \qquad \Phi(3) = \alpha. \tag{12}$$

If h, e are both even, (11) gives $2d = e - h$. Write $h = 2r$, $e = 2m$. Since d is even, $m \equiv r \pmod 2$. For even indices,

$$C(2s) = L_{2s} - 2 = \begin{cases} L_s^2, & s \text{ odd}, \\ 5F_s^2, & s \text{ even}. \end{cases}$$

Equation (10) gives the two quotient conditions in case 1.

If both gaps are odd, put $u = (e - h)/2$. Equation (11) becomes

$$\alpha^{2(d-u)} = \Phi(h)/\Phi(e).$$

When $h \geq 3$ this ratio lies strictly between 1 and $\Phi(3) = \alpha < \alpha^2$, which is impossible for an integral exponent $2(d - u)$. When $h = 1$, the ratio is at least $\Phi(1)/\Phi(3) = \alpha^2$, with equality at $e = 3$; for $e > 3$ it lies strictly between α^2 and $\alpha^3 < \alpha^4$. Therefore $h = 1, e = 3$; then $d = 2$ and (10) gives $q = 2$.

If h is odd and e is even, the extra factor in (11) is $\Phi(h)$. For odd $h \geq 5$, the right side lies strictly between consecutive powers α^{e-h} and α^{e-h+1}, of which only the second has an even exponent. This is impossible. For $h = 1$, (12) gives $2d = e + 2$; for $h = 3$ it gives $2d = e - 2$. Putting $e = 2m$ in the norm equation then gives, respectively, $q = L_m$ or $q = L_m/2$ when m is odd. Indeed m even would require an irrational positive multiple of $\sqrt{5}$ to be the integer q. In the second case, integrality of $L_m/2$ is equivalent to $3 \mid m$; the condition $q \geq 2$ also excludes $m = 1$.

Finally, for even h, odd e, (11) has the factor $1/\Phi(e)$. For $e \geq 5$, the exponent lies strictly between the consecutive integers $e - h - 1$ and $e - h$; the former is even, and no even exponent is possible. Thus $e = 3$, necessarily $h = 2$, and (12) gives $d = 0$ and $q = 2$.

For sufficiency, case 1 is the quotient identity (2). Cases 2 and 3 follow from $L_{n+2m} - L_n = L_m L_{n+m}$ for odd m, together with $L_{n+1} - L_n = L_{n-1}$ and $L_{n+3} - L_n = 2L_{n+1}$. The last two cases are the corresponding $q = 2$ identities. All hold for every sufficiently large common shift, with strictly increasing positive values. □

Theorem 6.2 (Explicit infinite-collision coefficient classification)**.** *For every integer $q \geq 2$, the set T_q has infinitely many multiply represented positive values if and only if q belongs to the following union:*

$$\boxed{\begin{aligned} &\{L_m/L_r \in \mathbb{Z} : m > r \geq 1,\ m, r \text{ odd}\} \\ \cup\ &\{F_m/F_r \in \mathbb{Z} : m > r \geq 2,\ m, r \text{ even}\} \\ \cup\ &\{L_{6r+3}/2 : r \geq 0\}. \end{aligned}} \tag{13}$$

The same equivalence holds for P_q. Every other integer $q \geq 2$ has only finitely many collisions in T_q and P_q.

Proof. Each coefficient in (13) has an infinite family of positive collision values by Theorems 2.1 and 2.2. Conversely, if T_q has infinitely many collision values, Theorem 5.1 forces a collision holding under every common index shift. Lemma 6.1 puts its coefficient in one of the three families: cases 2 and 4 are subsumed by the first or third family, and case 3 is the third family. The equivalence for P_q and finiteness outside these families follow from Proposition 5.2. □

7 Finite-collision coefficient spectra

The preceding global bounds also allow us to classify the coefficients with finitely many collisions without computing any ordered set of positive values. The sole arithmetic objects needed are differences of two Lucas numbers and the sign of a two-by-two determinant.

For integers $1 \le a < b$ put $D(a,b) = L_b - L_a > 0$. Given two index pairs $a < b$ and $c < d$, define

$$X = D(a,b), \quad Y = D(c,d), \qquad \Delta(a,b;c,d) = L_a L_d - L_b L_c.$$

The integrality condition $Y \mid X$ generates the candidate coefficient $q = X/Y$.

Lemma 7.1 (Uniqueness of positive Lucas ratios)**.** *For $a > b \ge 1$, $c > d \ge 1$, the equality $L_a/L_b = L_c/L_d$ implies $(a,b) = (c,d)$.*

Proof. Set $R_h(n) = L_{n+h}/L_n$, $\alpha = \varphi$. Binet's formula gives, for $n \ge 2, h \ge 1$,

$$\frac{4}{3\alpha}\alpha^h \le R_h(n) \le \frac{2}{\alpha}\alpha^h, \tag{14}$$

since

$$\frac{1-\alpha^{-6}}{1+\alpha^{-4}} = \frac{4}{3\alpha}, \qquad \frac{1+\alpha^{-6}}{1-\alpha^{-4}} = \frac{2}{\alpha}.$$

The intervals for distinct h are disjoint, because $4/3 > 2/\alpha$. For fixed h,

$$L_m L_{n+h} - L_n L_{m+h} = 5(-1)^{n+1} F_{m-n} F_h \ne 0 \qquad (m > n \ge 2).$$

Thus all ratios with $n \ge 2$ are distinct.

For $n = 1$, write $R_h(1) = L_s$, $s = h + 1 \ge 2$. Since

$$(1-\alpha^{-4})\alpha^s \le L_s \le (1+\alpha^{-4})\alpha^s,$$

the inequalities $\alpha(1-\alpha^{-4}) > 2/\alpha$ and $1 + \alpha^{-4} < 4/3$ show that $L_s = R_e(m)$, $m \ge 2$, would force $e = s$. But then the Lucas product formula gives

$$L_s L_m - L_{m+s} = (-1)^s L_{m-s} \ne 0,$$

including $m < s$ via the standard extension to negative Lucas indices. Ratios with both denominators L_1 are strictly increasing with their numerators. □

Theorem 7.2 (Signed-gap coefficient spectra)**.** *Fix $q \ge 2$. The following statements are exact necessary-and-sufficient conditions for the existence of a multiply represented* positive *value:*

1. *S_q has a collision if and only if some $1 \le a < b$, $1 \le c < d$ satisfy $D(a,b) = qD(c,d)$ and $\Delta(a,b;c,d) > 0$.*
2. *T_q has a collision if and only if such indices satisfy $\Delta(a,b;c,d) < 0$.*
3. *P_q has a collision if and only if such indices exist, regardless of the sign of Δ.*

All three tests can be restricted to the common finite index window

$$\boxed{1 \le a, b, c, d \le 13t + 1,} \tag{15}$$

where $t = t(q)$ is defined by (5). The two strict sign conditions on Δ characterize positive collisions in S_q and T_q.

Proof. If $D(a,b) = qD(c,d)$, then the two unequal pairs of indices give the sum collision

$$qL_c + L_b = qL_d + L_a > 0.$$

Furthermore,

$$L_a - qL_c = L_b - qL_d = \frac{\Delta}{D(c,d)},$$

and

$$qL_c - L_a = qL_d - L_b = -\frac{\Delta}{D(c,d)}.$$

This proves that positive, negative, and zero signs have precisely the asserted consequences. Conversely, any collision in S_q has first indices $a < b$ and second indices $c < d$; its equality and positivity give the divisor equation and $\Delta > 0$. For a T_q collision, label the first indices $c < d$ and second indices $a < b$; the same operations yield $\Delta < 0$. For P_q, the two first and second indices necessarily run in opposite directions; relabeling them gives the divisor equation. Thus all three tests are equivalent, not merely sufficient.

If $\Delta = 0$, the divisor equation implies $L_a = qL_c$ and $L_b = qL_d$. These are two distinct representations of the same Lucas ratio $q \geq 2$, contrary to Lemma 7.1. Thus no zero-determinant coefficient candidate exists.

For the common index bound, Theorem 4.1 places all S_q collision indices at most $6t$. If q has no stationary T_q collision, Proposition 5.2 places every T_q and P_q collision inside $13t$. Otherwise Theorem 5.1 supplies a positive stationary T_q collision at $k = 12t + 1$, whose other three indices belong to $[1, 13t + 1]$; exchanging subtracted indices also supplies a P_q collision inside that window. In all cases, if a collision exists, a certificate satisfying (15) exists. □

Define two explicitly parametrized sets of positive integer coefficients:

$$\boxed{\mathcal{C}_\pm = \left\{ \frac{D(a,b)}{D(c,d)} \in \mathbb{Z}_{\geq 2} : \ a < b, \ c < d, \ \pm\Delta(a,b;c,d) > 0 \right\}.} \tag{16}$$

All four indices are positive. Thus S_q has a positive collision precisely for $q \in \mathcal{C}_+$; T_q precisely for $q \in \mathcal{C}_-$; and P_q precisely for $q \in \mathcal{C}_+ \cup \mathcal{C}_-$. Let $\mathcal{I}$ be the three explicit coefficient families in (13). By Theorem 6.2, the coefficients admitting a finite, nonempty set of collisions are

$$\begin{array}{c|l} S_q & \mathcal{C}_+ \\ T_q & \mathcal{C}_- \setminus \mathcal{I} \\ P_q & (\mathcal{C}_+ \cup \mathcal{C}_-) \setminus \mathcal{I}. \end{array}$$

The complements give global uniqueness, while $\mathcal{I}$ gives infinitely many collision values for T_q and P_q. For S_q, all collisions are finite at every $q \geq 2$.

Corollary 7.3 (Sum uniqueness from both differences)**.** *For every integer $q \geq 2$, all positive values in P_q have unique index-pair representations if and only if this holds simultaneously in S_q and T_q.*

Proof. Theorem 7.2 partitions all positive P_q collision witnesses into the two strict signs, respectively equivalent to S_q and T_q collisions. The zero sign is impossible by Lemma 7.1. □

Proposition 7.4 (Equal-gap parity rule)**.** *For $h \geq 1$ and $1 \leq c < a$, suppose*

$$q = \frac{L_{a+h} - L_a}{L_{c+h} - L_c} \in \mathbb{Z}_{\geq 2}.$$

Then P_q has repeated positive values. If c is odd, S_q has a positive collision; if c is even, T_q has a positive collision. In particular, each of the infinite families

$$\boxed{q = \frac{L_{3r}}{2}\ (r \geq 1), \qquad q = \frac{L_{4r+2}}{3}\ (r \geq 1), \qquad q = \frac{F_{3r}}{2}\ (r \geq 2)}$$

consists entirely of coefficients for which S_q has a finite nonempty collision set.

Proof. The Lucas–Fibonacci product identity gives

$$\boxed{L_a L_{c+h} - L_c L_{a+h} = 5(-1)^{c+1} F_{a-c} F_h.} \tag{17}$$

For $a > c$ and $h \geq 1$, both Fibonacci factors are positive. Theorem 7.2 therefore proves the parity rule.

For the three families choose $c = 1$. Using the elementary identities $L_{n+1} - L_n = L_{n-1}$, $L_{n+2} - L_n = L_{n+1}$, and $L_{n+4} - L_n = 5F_{n+2}$, take

$$(h, a) = (1, 3r+1), \qquad (2, 4r+1), \qquad (4, 3r-2),$$

respectively (the last has $r \geq 2$). The denominators $D(1, 1+h)$ are $2, 3, 10$. Divisibility follows from the period-three Lucas/Fibonacci parities and the Lucas residues $L_{4r+2} \equiv 0 \pmod 3$. Theorem 4.1 proves that every fixed S_q has only finitely many collisions. □

8 A family with globally unique differences

Theorem 8.1. *Fix $r \geq 1$ and set $q = 5F_{2r}$. Then*

$$L_i - qL_j > 0 \quad \Longleftrightarrow \quad (i, j) = (2r+2, 1) \quad \text{or} \quad i \geq 2r+3,\ 1 \leq j \leq i - 2r - 2.$$

Every positive integer in S_q has precisely one representation.

Proof. The recurrence yields

$$q = 5F_{2r} = L_{2r+2} - L_{2r-2}, \quad q - L_{2r+1} = L_{2r-1} > 0, \quad L_{2r+2} - q = L_{2r-2} > 0. \tag{18}$$

Thus $j = 1$ first becomes admissible at $i = 2r+2$. For $j \geq 2$ put $A_j = L_{j+2r+2} - qL_j$ and $B_j = qL_j - L_{j+2r+1}$. Both obey the Lucas recurrence in j. Checking two initial values in r and using the recurrence in steps of two gives

$$A_2 = L_{2r+4} - 3q = L_{2r-4} > 0, \quad A_3 = A_2 + A_1 > 0, \quad A_1 = L_{2r+3} - q > 0,$$

with $L_{-2} = 3$ when $r = 1$. For B_j, the first two terms of the sequences $B_2(r)$ and $B_3(r)$ are

$$(B_2(1), B_2(2)) = (4, 16), \qquad (B_3(1), B_3(2)) = (2, 13).$$

Each satisfies $b_{r+2} = 3b_{r+1} - b_r$ as a function of r. An increasing positive initial pair remains increasing and positive. Therefore $B_2, B_3 > 0$ for all $r \geq 1$; the recurrence in j then gives $A_j, B_j > 0$ for all $j \geq 2$. Hence $L_{j+2r+1} < qL_j < L_{j+2r+2}$ for $j \geq 2$, establishing the stated index region.

Let $\mathcal{B}_i$ be the positive values with first index i. They are distinct within each block, and $\max \mathcal{B}_i = L_i - q$ for $i \geq 2r+2$. Using (1) with $b = i - 2r + 2$, the next block of the same residue class satisfies

$$\begin{aligned} \min \mathcal{B}_{i+4} - \max \mathcal{B}_i &= L_{i+4} - L_i - qL_{i-2r+2} + q \\ &= q + 5F_{i-4r+2} > 0. \end{aligned} \tag{19}$$

For a nonnegative Fibonacci index the inequality is immediate. For a negative index, $i \geq 2r+2$ implies $0 < |i-4r+2| \leq 2r-4 < 2r$, so $|F_{i-4r+2}| < F_{2r}$. Moreover $A_{j+4} > A_j$ follows from the strictly positive Lucas recurrence; thus blocks separated by any positive multiple of four are strictly ordered.

Finally, $(L_1, L_2, L_3, L_4) \bmod 5 = (1, 3, 4, 2)$ is the repeating residue cycle. Since $5 \mid q$, two equal differences must have first indices congruent modulo four. They cannot belong to different such blocks, by (19), and the same block has distinct values. □

9 The four OEIS sequences at coefficient five

The preceding classification applies in particular to $q = 5$. For this section write $S = S_5$, $T = T_5$, and $P = P_5$. Their increasing positive values give OEIS A191981, A191982, and A191980, respectively; the positive values of A191983 form $S \cup T$. The published A191981, A191982, and A191983 have 0 as their initial term. Because $L_i \bmod 5$ cycles through $1, 3, 4, 2$, the equation $L_i = 5L_j$ is impossible: the zero is not a Lucas difference.

Theorem 8.1 with $r = 1$ gives the complete index region and uniqueness in S:

$$(i, j) = (4, 1) \quad \text{or} \quad i \geq 5,\ 1 \leq j \leq i-4.$$

For $i \geq 5$ its ordered block satisfies

$$3L_{i-5} = L_i - 5L_{i-4} \leq L_i - 5L_j \leq L_i - 5. \tag{20}$$

Write $H(y) = \#\{j \geq 1 : L_j \leq y\}$ and $[x]_+ = \max(x, 0)$. The exact conclusions for the four sequences are as follows.

9.1 Exact positive-difference count

Theorem 9.1 (Three-block exact counting)**.** *For every integer $x \geq 11$, let m be the unique integer satisfying $L_m \leq x < L_{m+1}$. Then*

$$\boxed{A(x) = 1 + \frac{(m-4)(m-3)}{2} + \sum_{h=1}^{3} \left[m + h - 4 - H\left(\left\lfloor \frac{L_{m+h} - x - 1}{5} \right\rfloor \right) \right]_+ .} \tag{21}$$

For $0 \leq x < 11$, $A(x)$ counts the members of $\{2, 3, 6, 9\}$ not exceeding x.

Proof. All admissible differences with first index $i \leq m$ are smaller than $L_i \leq x$. For $i \geq m+4$, (20) gives

$$L_i - 5L_j \geq 3L_{i-5} \geq 3L_{m-1} > L_{m+1} > x.$$

Indeed, $3L_{m-1} - L_{m+1} = L_{m-1} - L_{m-2} > 0$ because $m \geq 5$. The only remaining first indices are $i = m+1, m+2, m+3$. In one such block, $L_i - 5L_j \leq x$ is equivalent to $L_j > \lfloor (L_i - x - 1)/5 \rfloor$. There are $i-4$ admissible indices j, so removing the ineligible ones gives the positive-part expression in (21). □

9.2 Complete intersection of the two differences

Theorem 9.2 (Complete intersection classification)**.** *With $E = \{2, 6, 9, 12, 13, 14, 24, 32, 61\}$, one has the disjoint union*

$$\boxed{S \cap T = E \,\dot\cup\, \{3L_n : n \geq 6\} \,\dot\cup\, \{8L_n : n \geq 6\}.} \tag{22}$$

In particular, the only common values outside the two Lucas-multiple families are the nine elements of E.

Proof. Suppose $L_i - 5L_j = 5L_k - L_\ell > 0$. The Lucas residues modulo 5 repeat as $(1, 3, 4, 2)$, so $L_i + L_\ell \equiv 0 \pmod 5$ forces $\ell \equiv i + 2 \pmod 4$. By Theorem 8.1, $i \geq 4$ and $j \leq i - 4$, except that $(i, j) = (4, 1)$ is the first permitted pair. Furthermore, $5L_k > L_\ell$ implies $\ell \leq k + 3$ for every $k \geq 1$. For $k \geq 2$, the least positive difference in its block is $5L_k - L_{k+3} = 2L_{k-2}$.

Assume first $i \geq 16$. The lower bound $L_i - 5L_j \geq 3L_{i-5} > 5L_{i-7}$ excludes $k \leq i - 7$. The bound $5L_k - L_\ell \geq 2L_{k-2} \geq 2L_{i-1} > L_i$ excludes $k \geq i + 1$. If $k = i$, the largest permissible ℓ is $i + 2$ and $5L_i - L_{i+2} > L_i$. If $k = i - 2$, the largest permissible ℓ is $i - 2$ and $4L_{i-2} > L_i$. Thus $k = i - d$ with $d \in \{1, 3, 4, 5, 6\}$. For $d = 1$, only $\ell = i + 2$ can give a value below L_i; all smaller permissible ℓ give $5L_{i-1} - L_\ell > L_i$.

Put $R = L_i + L_\ell - 5L_{i-d} = 5L_j$. The following table gives strict open intervals for R; each endpoint is a multiple of a Lucas number.

d	ℓ	location of R
1	$i + 2$	$(5L_{i-5},\ 5L_{i-4})$
3	$i - 2$	$(5L_{i-7},\ 5L_{i-6})$
3	$\ell \leq i - 6$	$R < 0$
4	$i - 2$	$(5L_{i-5},\ 5L_{i-4})$
4	$i - 6$	$(5L_{i-6},\ 5L_{i-5})$
4	$i - 10$	$R = 5L_{i-6}$
4	$\ell \leq i - 14$	$(5L_{i-7},\ 5L_{i-6})$
5	$i - 2$	$R > 5L_{i-4}$
5	$\ell \leq i - 6$	$(5L_{i-5},\ 5L_{i-4})$
6	$i - 6$	$R > 5L_{i-4}$
6	$i - 10$	$R = 5L_{i-4}$
6	$\ell \leq i - 14$	$(5L_{i-5},\ 5L_{i-4})$

All table inequalities hold for $i \geq 16$. To verify them, bound each variable ℓ between its indicated upper endpoint and $L_\ell > 0$. The resulting fixed-shift inequalities are differences of Lucas-linear expressions and hence satisfy $u_{i+2} = u_{i+1} + u_i$. In each case the two initial values $i = 16, 17$ are strictly positive, as checked by direct substitution; the recurrence preserves positivity. The two exact-equality rows follow from the same recurrence and two initial identities. Every interval either misses all admissible $5L_j$ or contains no such value, while $R > 5L_{i-4}$ is impossible since $j \leq i - 4$. The only surviving cases are

$$(i, j, k, \ell) = (i, i - 6, i - 4, i - 10), \qquad (i, i - 4, i - 6, i - 10) \qquad (i \geq 16).$$

For the remaining $4 \leq i \leq 15$, the finite possibilities $1 \leq k \leq i - 1$, $1 \leq \ell \leq k + 3$, and $j \leq i - 4$ (with $(4, 1)$ included) give:

i	4	5	6	7	8	9	10
$S \cap T$ in the i-block	2	6	13	$9, 14, 24$	$12, 32$	61	$\varnothing$

For $11 \leq i \leq 15$ the only two values are $3L_{i-5}$ and $8L_{i-5}$. For all $n \geq 6$ the Lucas recurrence gives

$$3L_n = L_{n+5} - 5L_{n+1} = 5L_{n-1} - L_{n-5},$$
$$8L_n = L_{n+5} - 5L_{n-1} = 5L_{n+1} - L_{n-5}.$$

Thus both displayed families belong to $S \cap T$. They are disjoint from one another by the uniqueness theorem for S, and neither family contains one of the nine exceptional values (the smallest values are $3L_6 = 54$ and $8L_6 = 144$). This proves (22). □

9.3 All repeated reverse differences and sums

Theorem 9.3 (Five and only five collisions)**.** *The map*

$$(k,\ell) \longmapsto 5L_k - L_\ell > 0, \qquad k,\ell \geq 1,$$

is injective except for exactly five pairs of distinct arguments:

k	p	(ℓ, q)	$5L_k - L_\ell = 5L_p - L_q$
1	2	$(1,5)$	4
1	3	$(2,6)$	2
2	5	$(4,8)$	8
2	6	$(1,9)$	14
3	4	$(2,6)$	17

In particular, the set of multiply represented positive values in A191982 is precisely $D = \{2,4,8,14,17\}$*; each has multiplicity two.*

Proof. For $k = 1$ the positive pairs have $1 \leq \ell \leq 3$. For every $k \geq 2$, positivity is equivalent to $1 \leq \ell \leq k+3$, and the values in the k-block lie in

$$[2L_{k-2},\, 5L_k - 1], \qquad 5L_k - L_{k+3} = 2L_{k-2}.$$

Suppose $5L_k - L_\ell = 5L_p - L_q > 0$ with $k < p$. Then

$$L_q - L_\ell = 5(L_p - L_k) > 0, \qquad q \equiv \ell \pmod 4. \tag{23}$$

The congruence follows from the four distinct Lucas residues modulo 5. Hence $q \geq \ell + 4$.

If $k \geq 3$ and $p \geq k+4$, then

$$5L_p - L_q \geq 2L_{p-2} \geq 2L_{k+2} > 5L_k - 1,$$

where $2L_{k+2} - 5L_k = L_{k-1} - L_{k-2} > 0$. This exceeds every value in the k-block. For $k = 1$, $p \geq 5$ is excluded by $2L_3 = 8 > 5L_1 - 1 = 4$. For $k = 2$, $p \geq 7$ is excluded by $2L_5 = 22 > 5L_2 - 1 = 14$; the boundary $p = 6$ gives precisely 14.

It remains to consider $p = k + d$ for $d \in \{1,2,3\}$. For $k \geq 4$ the possibilities in (23) reduce as follows.

If $d = 1$, then $5(L_p - L_k) = 5L_{k-1}$, so $L_{k+2} < 5L_{k-1} < L_{k+4}$ and $q \in \{k+3, k+4\}$. For $q = k+3$ one has $L_\ell = 3L_{k-2} > L_{k-1}$, contradicting $\ell \leq q - 4 = k - 1$. For $q = k+4$ one has $L_\ell = L_{k+2} + 3L_{k-2} > L_k$, contradicting $\ell \leq k$.

If $d = 2$, then $5(L_p - L_k) = 5L_{k+1}$. Since $L_{k+4} < 5L_{k+1}$ and $q \leq p+3$, necessarily $q = k+5$. It follows that $L_\ell = 3L_k > L_{k+1}$, but $\ell \leq k+1$.

If $d = 3$, then $5(L_p - L_k) = 10L_{k+1}$. As $L_{k+5} < 10L_{k+1}$ and $q \leq p+3$, one must have $q = k+6$. Thus

$$L_\ell = L_{k+6} - 10L_{k+1} = 5L_k - 2L_{k+1} =: R_k.$$

For $k \geq 3$,

$$L_{k-2} < R_k < L_{k+2}.$$

But (23) forces $\ell \equiv k+2 \pmod 4$ and $\ell \leq k+2$; either $\ell = k+2$ or $\ell \leq k-2$, a contradiction.

Finally, the cases $k = 1,2,3$ can be checked using (23) and $q \leq p+3$: $k = 1$ yields $(p,\ell,q) = (2,1,5), (3,2,6)$; $k = 2$ yields $(5,4,8)$ in addition to the boundary $(6,1,9)$; $k = 3$ yields $(4,2,6)$. Their values are the five displayed in the table. Since the list exhausts all colliding argument pairs, no value has three representations. □

Theorem 9.4 (The five sum collisions)**.** *Exactly five values in P admit two different index-pair representations:*

s	*the two representations of* s
16	$5L_1 + L_5 = 5L_2 + L_1$
23	$5L_1 + L_6 = 5L_3 + L_2$
38	$5L_3 + L_6 = 5L_4 + L_2$
62	$5L_2 + L_8 = 5L_5 + L_4$
91	$5L_2 + L_9 = 5L_6 + L_1$

All other values have precisely one representation; none has three.

Proof. Suppose $5L_i + L_j = 5L_k + L_\ell$ with $(i, j) \neq (k, \ell)$. Since L_n increases strictly for $n \geq 1$, we may take $i < k$ and $j > \ell$. Rearranging gives

$$5L_i - L_\ell = 5L_k - L_j =: t. \tag{24}$$

The four Lucas residues $1, 3, 4, 2$ modulo 5 imply $t \neq 0$. If $t < 0$, then $-t = L_\ell - 5L_i = L_j - 5L_k > 0$ has two distinct representations in S, contradicting Theorem 8.1. Consequently $t > 0$. Equation (24) is therefore a collision in T, and Theorem 9.3 restricts it to the five pairs listed there. Conversely, swapping the subtracted Lucas indices in each of those five equalities yields exactly the five sum identities displayed above. Each displayed sum is distinct, and each of the five collisions in T is between precisely two index pairs; hence there are no further collisions and no threefold representations in P. □

9.4 Complete intersection of sums and reverse differences

Theorem 9.5 (Complete sum–difference intersection)**.** *Put*

$$E_{PT} = \{6, 8, 9, 12, 16, 19, 22, 24, 26, 31, 34, 44, 52, 138\}.$$

The following union is disjoint:

$$\boxed{P \cap T = E_{PT} \,\dot\cup\, \{5L_n + L_{n+7} : n \geq 4\} \,\dot\cup\, \{5L_n + L_{n-3} : n \geq 4\}.} \tag{25}$$

Indeed, every equality $5L_a + L_b = 5L_c - L_d > 0$ with $c \geq 8$ has $a = c - 4$ and $\{b, d\} = \{c + 3, c - 7\}$.

Proof. The equality implies $a < c$, since $L_b + L_d > 0$, and

$$L_b + L_d = 5(L_c - L_a), \qquad b \equiv d + 2 \pmod 4. \tag{26}$$

The residue condition follows from the four distinct Lucas residues modulo 5. Positivity gives $d \leq c + 3$; also $L_b < 5L_c < L_{c+4}$ gives $b \leq c + 3$.

Assume $c \geq 8$, and set $m = \max(b, d)$ and $s = \min(b, d)$. The residue condition forces $m - s \equiv 2 \pmod 4$, so $s \leq m - 2$. If $m = c + 3$, then $s = c + 1$ would make $L_m + L_s > 5L_c$, hence $s \leq c - 3$. If $m = c + 2$, then $s \leq c$; if $m \leq c + 1$, then $s \leq c - 1$. The Lucas recurrence yields

$$L_m + L_s \leq L_{c+3} + L_{c-3}, \qquad 5L_c - L_{c+3} - L_{c-3} > 5L_{c-5}.$$

Thus $L_a > L_{c-5}$, hence $a \geq c - 4$. Write $h = c - a \in \{1, 2, 3, 4\}$ and $R_h = 5(L_c - L_{c-h}) = L_m + L_s$.

For $h = 1$, the bounds $2L_{c-1} < R_1 < L_{c+2}$ force $m = c$ or $c + 1$. When $m = c$, $R_1 > L_c + L_{c-2}$ contradicts $s \leq c - 2$. When $m = c + 1$,

$$L_{c-5} < R_1 - L_{c+1} < L_{c-1}.$$

But then s must belong to $\{c-1, c-5, c-9, \ldots\}$, which is impossible by strict monotonicity. For $h = 2$, the bounds $2L_c < R_2 < L_{c+3}$ force $m = c+1$ or $c+2$. The former is excluded by $R_2 > L_{c+1} + L_{c-1}$; in the latter case

$$L_{c-4} < R_2 - L_{c+2} < L_c,$$

whereas $s \in \{c, c-4, c-8, \ldots\}$. For $h = 3$, $2L_{c+1} < R_3 < L_{c+3}$ forces $m = c+2$, but $R_3 > L_{c+2} + L_c$ contradicts $s \le c$. Finally, for $h = 4$, $R_4 > L_{c+2} + L_c$ forces $m = c+3$; the identity

$$5(L_c - L_{c-4}) = L_{c+3} + L_{c-7}$$

gives $s = c - 7$. This gives precisely the two families in (25).

All strict inequalities used above are fixed-shift linear combinations of Lucas numbers. Their differences satisfy $u_{c+2} = u_{c+1} + u_c$; each is positive at $c = 8, 9$, so all hold for $c \ge 8$. The 18 initial comparisons at $c = 8, 9$ and the exact identity follow by direct substitution.

For $c \le 7$, the ranges $1 \le a < c$ and $1 \le b, d \le c+3$ are finite. Exhaustion gives the following distinct values, grouped by c:

c	$5L_a + L_b = 5L_c - L_d$ (distinct values)
1	$\varnothing$
2	$8, 12$
3	$9, 16, 19$
4	$6, 24, 31, 34$
5	$8, 26, 44, 52$
6	$\varnothing$
7	$22, 138$

These give exactly E_{PT}. The family with L_{n+7} begins at $234 > \max E_{PT}$; the other begins with $36, 58, 94, 152, \ldots$, none of which belongs to E_{PT}. The two families are disjoint by Theorem 9.4: if one value belonged to both, it would have two different sum representations, yet every value of the first family exceeds 91. The Lucas identity

$$5(L_{n+4} - L_n) = L_{n+7} + L_{n-3} \qquad (n \ge 4)$$

holds by the recurrence and its first two instances. It gives both families in $P \cap T$; the second difference is positive since $5L_{n+4} - L_{n+7} = 2L_{n+2} > 0$. □

9.5 Exact counting and unranking

Let

$$A(x) = \#(S \cap [1, x]), \quad B(x) = \#(T \cap [1, x]), \quad C(x) = \#(P \cap [1, x]).$$

The first count is (21) for $x \ge 11$; for $0 \le x < 11$, count the elements of $\{2, 3, 6, 9\}$ at most x. Let

$$D_T = \{2, 4, 8, 14, 17\}, \qquad D_P = \{16, 23, 38, 62, 91\},$$

and put $b_1 = 3$, $b_k = k + 3$ for $k \ge 2$, $K(x) = \max\{3, H(\lfloor x/2 \rfloor) + 2\}$. The two remaining count formulas are

$$B(x) = \sum_{k=1}^{K(x)} [b_k - H(5L_k - x - 1)]_+ - \#\{d \in D_T : d \le x\}, \tag{27}$$

$$C(x) = \sum_{i=1}^{H(\lfloor (x-1)/5 \rfloor)} H(x - 5L_i) - \#\{d \in D_P : d \le x\}. \tag{28}$$

For the two intersections let $I(x) = \#(S \cap T \cap [1, x])$ and $J(x) = \#(P \cap T \cap [1, x])$. Write

$$E_{ST} = \{2, 6, 9, 12, 13, 14, 24, 32, 61\},$$

and $u_n = 5L_n + L_{n+7}$, $v_n = 5L_n + L_{n-3}$ for $n \geq 4$. Then the two intersection theorems yield

$$I(x) = \#\{e \in E_{ST} : e \leq x\} + [H(\lfloor x/3 \rfloor) - 5]_+ + [H(\lfloor x/8 \rfloor) - 5]_+, \tag{29}$$

$$J(x) = \#\{e \in E_{PT} : e \leq x\} + \#\{n \geq 4 : u_n \leq x\} + \#\{n \geq 4 : v_n \leq x\}. \tag{30}$$

Finally, for $x \geq 0$, the number of entries of A191983 not exceeding x is

$$\boxed{U(x) = 1 + A(x) + B(x) - I(x).} \tag{31}$$

Every sum in these formulas is finite. They count pairs within their permitted Lucas index regions, then subtract the five duplicated values in T or P, respectively. The intersection counts follow from the disjoint unions proved above, and U follows from inclusion–exclusion. All four ordered sequences and both intersections therefore admit exact unranking by integer binary search.

9.6 Direct terms of the four OEIS sequences

Write $a_{980}(n), a_{981}(n), a_{982}(n), a_{983}(n)$ for the published sequences, indexed from $n = 1$. The sequences $a_{981}, a_{982}, a_{983}$ start with zero.

Theorem 9.6 (All-index direct expressions)**.** *For $n \geq 27$, define*

$$k = \left\lfloor \frac{1 + \sqrt{8n - 41}}{2} \right\rfloor, \qquad r = n - 5 - \frac{k(k-1)}{2}.$$

Then $1 \leq r \leq k$ and

$$a_{981}(n) = \begin{cases} 3L_{k+1}, & r = 1, \\ L_{k+4} - 5L_{k-r}, & r = 2, 3, \\ L_{k+5} - 5L_k, & r = 4, \\ L_{k+4} - 5L_{k+1-r}, & 5 \leq r \leq k. \end{cases} \tag{32}$$

For $n \geq 15$, define

$$m = \left\lfloor \frac{\sqrt{8n + 49} - 5}{2} \right\rfloor, \qquad r = n - \frac{m^2 + 5m - 8}{2}.$$

Then $1 \leq r \leq m + 3$ and

$$a_{982}(n) = \begin{cases} 2L_{m+1}, & r = 1, \\ 5L_m - L_{m+1}, & r = 2, \\ 5L_{m+1} - L_{m+3}, & r = 3, \\ 5L_m - L_{m+4-r}, & 4 \leq r \leq m + 3. \end{cases} \tag{33}$$

For $n \geq 81$, define

$$m = \left\lfloor \frac{5 + \sqrt{4n + 37}}{2} \right\rfloor, \qquad r = n - m^2 + 5m + 4.$$

Then $1 \leq r \leq 2m - 4$ and

$$a_{980}(n) = 5L_{i(r)} + L_{j(r)}, \tag{34}$$

where

r	$i(r)$	$j(r)$
$1 \le r \le m-9$	r	m
$m-8$	$m-5$	$m-1$
$m-7$	$m-8$	m
$m-6$	$m-4$	$m-2$
$m-5$	$m-7$	m
$m-4 \le r \le 2m-10$	$m-3$	$r-m+5$
$2m-9$	$m-6$	m
$2m-8$	$m-3$	$m-4$
$2m-7$	$m-4$	$m-1$
$2m-6$	$m-3$	$m-3$
$2m-5$	$m-5$	m
$2m-4$	$m-3$	$m-2$

For $n \ge 57$, define

$$m = 3 + \lfloor \sqrt{n-8} \rfloor, \qquad r = n - \big((m-3)^2 + 7\big).$$

Then $1 \le r \le 2m-5$, and

$$a_{983}(n) = \begin{cases} 5L_{m-3} - L_{m-3-r}, & 1 \le r \le m-4, \\ 2L_{m-1}, & r = m-3, \\ 5L_{m-2} - L_{m-1}, & r = m-2, \\ L_{m+1} - 5L_{m-6}, & r = m-1, \\ L_{m+2} - 5L_{m-3}, & r = m, \\ L_{m+1} - 5L_{m-7}, & r = m+1, \\ 5L_{m-1} - L_{m+1}, & r = m+2, \\ L_{m+1} - 5L_{m-8}, & r = m+3, \\ 4L_{m-2}, & r = m+4, \\ L_{m+1} - 5L_{2m-4-r}, & m+5 \le r \le 2m-5. \end{cases} \tag{35}$$

The finite prefixes below complete the formulas for every $n \ge 1$.

Finite initial terms. The lengths are $26, 14, 80, 56$, respectively.

A191981. $0, 2, 3, 6, 9, 12, 13, 14, 21, 24, 27, 32, 33, 41, 42, 54, 56, 61, 68, 71, 87, 88, 103, 108, 109, 118$.

A191982. $0, 1, 2, 4, 6, 8, 9, 11, 12, 13, 14, 16, 17, 19$.

A191980. $6, 8, 9, 12, 16, 18, 19, 21, 22, 23, 24, 26, 27, 31, 33, 34, 36, 38, 39, 42, 44, 46, 49, 52, 53, 56, 58, 59, 62, 64, 66, 67, 73, 81, 82, 84, 91, 93, 94, 96, 97, 101, 102, 108, 111, 119, 128, 131, 137, 138, 143, 146, 148, 149, 152, 156, 158, 163, 166, 174, 178, 192, 204, 213, 214, 219, 221, 234, 236, 238, 239, 242, 246, 253, 254, 264, 268, 282, 289, 311$.

A191983. $0, 1, 2, 3, 4, 6, 8, 9, 11, 12, 13, 14, 16, 17, 19, 21, 22, 24, 26, 27, 28, 31, 32, 33, 34, 36, 37, 41, 42, 43, 44, 48, 51, 52, 54, 56, 58, 61, 68, 69, 71, 72, 79, 83, 86, 87, 88, 89, 94, 98, 103, 108, 109, 112, 116, 118$.

Proof of Theorem 9.6. Partition the positive values into consecutive Lucas-scale intervals, enumerate their representations, and establish their strict order. All boundary and adjacent-order inequalities used below are fixed linear combinations of shifted Lucas numbers. Such combinations satisfy the Lucas recurrence, so two consecutive positive initial values imply positivity for every subsequent index.

For S, admissible pairs are $(4,1)$, or $i \geq 5,\ 1 \leq j \leq i-4$. For $m \geq 11$, the pairs in $[L_{m-1}, L_m)$, in increasing order, are

$$(m+2, m-2),\ (m, m-6),\ (m, m-7),\ (m+1, m-4),\ (m, m-8), \ldots, (m, 1).$$

Here (i,j) denotes $L_i - 5L_j$. To check the critical order and the left endpoint, put $x = L_{m-8}$, $y = L_{m-7}$. The first five gaps are

$$x + 2y, \quad y - x, \quad 5x, \quad 3y - 2x, \quad 2y - 3x,$$

and are positive for $m \geq 11$. The last inequality follows from the recurrence for $2L_{t+1} - 3L_t$, positive at $t = 3, 4$. The remaining row bounds follow from $L_i - 5L_{i-4} = 3L_{i-5}$. There are $m-4$ distinct terms per interval, and 26 terms below L_{10}, including zero. With $k = m-4$, the local rank r gives $n = 5 + k(k-1)/2 + r$, yielding (32).

For T, positive pairs are $1 \leq j \leq 3$ when $i = 1$, and $1 \leq j \leq i+3$ for $i \geq 2$. For $m \geq 4$, all values in $(5L_{m-1}, 5L_m]$ are, in increasing order,

$$(m+3, m+6),\ (m, m+1),\ (m+1, m+3),\ (m, m),\ (m, m-1), \ldots, (m, 1).$$

The bound $5L_i - L_{i+3} = 2L_{i-2}$ restricts the possible first indices to $m, \ldots, m+3$. The eleven endpoint and sorting inequalities are positive at $m = 4, 5$, hence for all $m \geq 4$. The interval has $m+3$ distinct terms. The initial 14 terms are at most 20; the last index through interval m is $(m^2 + 7m - 2)/2$. Its quadratic inverse gives (33).

For P, every interval $[L_m, L_{m+1})$ with $m \geq 12$ contains precisely the pairs

$$(i, m), \quad 1 \leq i \leq m-5, \qquad (m-3, j), \quad 1 \leq j \leq m-2,$$
$$(m-5, m-1),\ (m-4, m-2),\ (m-4, m-1).$$

The inequalities $i \leq m-3$, $j \leq m$ and the boundary comparisons at $m = 12, 13$ establish this list. The twelve lines of the table in (34) give the strict order; its adjacent differences are positive at $m = 12, 13$ and propagate by the recurrence. The interval contains $2m-4$ distinct values, and the 80 values below L_{12} are exactly the finite prefix. The cumulative count through interval m is $m^2 - 3m - 8$, yielding (34).

For $S \cup T$, every interval $[L_m, L_{m+1})$ with $m \geq 10$ contains the T-pairs

i	j
$m-3$	$1, \ldots, m-4$
$m-2$	$m-2, m-1$
$m-1$	$m+1$
m	$\varnothing$
$m+1$	$m+4$

and the S-pairs, in their own increasing order,

$$(m+3, m-1),\ (m+1, m-5),\ (m+1, m-6),\ (m+2, m-3),\ (m+1, m-7), \ldots, (m+1, 1).$$

Exactly two values overlap:

$$S(m+3, m-1) = T(m-3, m-7), \quad S(m+1, m-5) = T(m-3, m-9).$$

The ordered union consists of a T-run of $m-4$ terms, eight fixed intervening values, and an S-run of $m-9$ terms. Its endpoint and sorting inequalities hold at $m = 10, 11$, hence at every later m. The ten formula cases in (35) list exactly this order. There are $2m-5$ terms per interval, and 56 initial terms below L_{10}; hence the count below L_m is $(m-3)^2+7$. Inverting it yields (35). The remaining fixed-shift inequalities and the initial cases $m = 10, 11$ follow by direct substitution. □

Theorem 9.7 (Counting and term growth)**.** *With φ as defined in the introduction, as $x \to \infty$,*

$$A(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O(\log x), \qquad B(x) = \tfrac{1}{2}(\log_\varphi x)^2 + O(\log x),$$

$$C(x) = (\log_\varphi x)^2 + O(\log x), \qquad U(x) = (\log_\varphi x)^2 + O(\log x).$$

Consequently, as $n \to \infty$,

$$\log_\varphi a_{981}(n) = \sqrt{2n} + O(1), \qquad \log_\varphi a_{982}(n) = \sqrt{2n} + O(1),$$

$$\log_\varphi a_{980}(n) = \sqrt{n} + O(1), \qquad \log_\varphi a_{983}(n) = \sqrt{n} + O(1).$$

Proof. Put $M = \lfloor \log_\varphi x \rfloor$. The bounds $L_i \asymp \varphi^i$ give the following uniform index cutoffs, with absolute constants: every admissible S-pair with first index $i \le M - O(1)$ has value at most x, while a positive S-value at most x has $i \le M + O(1)$. The latter uses $L_i - 5L_j \ge 3L_{i-5}$ for every admissible $i \ge 5$. Summing the $i-4$ admissible pairs per block and using uniqueness yields $A(x) = M^2/2 + O(M)$.

Likewise all positive T-pairs with $k \le M - O(1)$ have value at most x; every such value at most x has $k \le M + O(1)$, since the minimum in a k-block for $k \ge 2$ is $2L_{k-2}$. Summing the $k+3$ admissible pairs and subtracting the five global duplications gives $B(x) = M^2/2 + O(M)$.

For P, all pairs with both indices at most $M - O(1)$ have sum at most x, and every pair producing a sum at most x has both indices at most $M + O(1)$. The five duplications thus give $C(x) = M^2 + O(M)$. The exact formula for $S \cap T$ consists of nine exceptional values and two Lucas families, so $I(x) = O(M)$. Inclusion–exclusion gives $U(x) = M^2 + O(M)$. Finally n differs by at most one from the corresponding positive counting function at its n-th term. Inverting these quadratic estimates yields the four asserted term asymptotics. □

Acknowledgments

Large language models, including OpenAI's ChatGPT, assisted with mathematical exploration, code development and checking, and manuscript drafting and editing. The authors directed and reviewed these contributions and take responsibility for the mathematical arguments and the final manuscript.

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